

37) What is System R? How many of its two major subsystems?

System R was designed and developed from 1974 to 1979 at IBM San Jose Research Centre. System R is the first implementation of SQL, which is the standard relational data query language, and it was also the first to demonstrate that RDBMS could provide better transaction processing performance. It is a prototype which is formed to show that it is possible to build a Relational System that can be used in a real-life environment to solve real-life problems.

Following are two major subsystems of System R:

- Research Storage
- System Relational Data System

38) What is Data Independence?

Data independence specifies that "the application is independent of the storage structure and access strategy of data." It makes you able to modify the schema definition at one level without altering the schema definition in the next higher level.

It makes you able to modify the schema definition in one level should not affect the schema definition in the next higher level.

There are two types of Data Independence:

Physical Data Independence: Physical data is the data stored in the database. It is in the bit-format. Modification in physical level should not affect the logical level.

For example: If we want to manipulate the data inside any table that should not change the format of the table.

Logical Data Independence: Logical data is the data about the database. It basically defines the structure. Such as tables stored in the database. Modification in logical level should not affect the view level.

For example: If we need to modify the format of any table, that modification should not affect the data inside it.